

From Fog Chamber to Aircraft Window: Pixel-Registered Imaging and Synthetic Fine-Tuning Enable Cross-Domain Defogging: supplemental document

S1. DATASET ROLES AND SPLIT PROVENANCE

Table S1 summarizes dataset roles and split provenance. The fog-chamber task is the only cross-architecture benchmark. The synthetic-fine-tuned, mixed O-HAZE+NH-HAZE, and NTIRE nighttime haze models are NAFNet fine-tuning regimes started from the same fog-chamber model. The synthetic-fine-tuned model is evaluated as direct transfer on fog-machine, aircraft-window, and public paired-haze examples; the public paired-haze and NTIRE task-specific models use supervised fine-tuning on their own training splits.

Table S1. Dataset roles, splits, and reported use.

Dataset	Train/validation use	Test/evaluation use	Reported role
Fog chamber benchmark	4943 paired display / target images	552 held-out paired images	cross-architecture restoration benchmark and NAFNet starting model
Fog chamber classification	3708 train, 412 validation clear-target images after excluding illustrations	460 held-out matched clear, foggy, and restored images	semantic-preservation check
Synthetic outdoor fog	18,000 Mapillary train, 2,000 validation, and 5,000 test clear images rendered on the fly with spatial synthetic fog	synthetic Mapillary test diagnostics plus direct-transfer examples on public haze, fog-machine, and aircraft-window images	synthetic fine-tuning
Mixed O-HAZE+NH-HAZE	O-HAZE 1–35 + NH-HAZE 1–45 train; O-HAZE 36–40 + NH-HAZE 46–50 validation	O-HAZE 41–45 + NH-HAZE 51–55 test	mixed real-haze fine-tuning
NTIRE 2026 nighttime haze	IDs 01–20 train, 21–23 validation	IDs 24–25 internal test	nighttime haze adaptation check
Fog-machine / aircraft	none; no aligned clear targets	qualitative examples only	out-of-distribution visual test

S2. METRICS

Four full-reference restoration metrics were calculated over the paired test sets: mean absolute error (MAE), mean squared error (MSE), peak signal-to-noise ratio (PSNR), and structural similarity index (SSIM) [1]. For an image with N pixels, ground-truth pixel values y , and reconstructed pixel values $\hat{y}$, the metrics are

$$\text{MAE} = \frac{1}{N} \sum_i |y_i - \hat{y}_i|, \quad (\text{S1})$$

$$\text{MSE} = \frac{1}{N} \sum_i (y_i - \hat{y}_i)^2, \quad (\text{S2})$$

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right), \quad (\text{S3})$$

$$\text{SSIM} = \frac{(2\mu_y\mu_{\hat{y}} + C_1)(2\sigma_{y\hat{y}} + C_2)}{(\mu_y^2 + \mu_{\hat{y}}^2 + C_1)(\sigma_y^2 + \sigma_{\hat{y}}^2 + C_2)}. \quad (\text{S4})$$

The means μ_y and $\mu_{\hat{y}}$, variances σ_y^2 and $\sigma_{\hat{y}}^2$, and cross-covariance $\sigma_{y\hat{y}}$ are computed between the paired ground-truth and reconstructed images. Pixel values were normalized to $[0, 1]$, so $\text{MAX} = 1$. The SSIM constants C_1 and C_2 are numerical stability constants; with the normalized data range used here they are $C_1 = (0.01)^2$ and $C_2 = (0.03)^2$.

For paired fog-imager restoration experiments we report mean per-image MAE, MSE, PSNR, and SSIM on held-out fog-chamber test images. PSNR was computed independently for each image and then averaged across the test set, so the reported mean PSNR is not necessarily the direct transform of the reported mean MSE. For synthetic Mapillary fog, validation and test PSNR are training diagnostics against the original clear crops and are not interpreted as real-fog transfer scores. Fog-machine and aircraft-window examples are qualitative tests because no paired clear targets exist. For O-HAZE, NH-HAZE, and NTIRE we report internal test-split metrics for supervised task-specific adaptation where clear targets are available. Classification metrics are reported separately below.

S3. FOG-CHAMBER MODEL BENCHMARK

The architecture benchmark used the vertical-polarizer medium-fog aligned dataset and the same archive clear images as targets. Samples were matched by exact filename within each category after sorting filenames. The split was deterministic within each category: every 10th paired image was held out for testing and all other matched pairs were used for training. This produced 4943 training pairs and 552 test pairs across apparel, artwork, cars, dishes, furniture, and illustrations.

Training used L1 loss, Adam with learning rate 10^{-4} , batch size 1, automatic mixed precision on CUDA, and a fixed 50-epoch budget for consistency across architectures rather than validation-based per-model stopping. Images were resized to 512×512 for training and evaluation. Memory-heavy models used training crops in the benchmark runner. Evaluation used full 512×512 images and reported mean per-image MAE, MSE, PSNR, and SSIM, see Table S2.

Table S2. Fog-imager restoration benchmark on the vertical-polarizer medium-fog test split. All rows appear in the bubble chart, completed 50 training epochs, and were evaluated on 552 held-out images. Train crop reports the crop size used for memory-heavy training; full indicates full 512×512 training images.

Model	PSNR	SSIM	MAE	MSE	Params (M)	Train crop
NAFNet [2]	24.33	0.7912	0.05046	0.00714	29.16	full
SpecAT S2 [3]	23.04	0.7508	0.05644	0.00888	0.99	full
RegGAN [4]	22.54	0.7447	0.05952	0.00917	11.38	full
ConvNeXt [5]	22.44	0.7188	0.06216	0.01061	29.52	full
RetinexFormer [6]	21.80	0.7298	0.06409	0.01013	1.61	full
HRNet [7]	21.67	0.7334	0.06632	0.01033	8.18	full
SpecAT S1 [3]	21.39	0.7028	0.06640	0.01046	0.29	full
MST++ [8]	19.96	0.6611	0.07879	0.01356	0.13	full
HDNet [9]	19.77	0.6760	0.08160	0.01438	2.34	full
PADUT [10]	18.98	0.6342	0.08981	0.01678	0.43	full
DehazeFormer [11]	18.86	0.6498	0.08616	0.01652	0.69	256
Swin2SR [12]	18.37	0.6505	0.09215	0.01878	1.00	256
RDN [13]	18.19	0.6205	0.09769	0.01966	2.23	full
Restormer [14]	17.63	0.6688	0.09425	0.02136	26.13	256
VmambaIR [15]	17.58	0.5743	0.10576	0.02251	1.35	full
MIRNet [16]	17.13	0.6268	0.10715	0.02494	31.79	128
MSBDN [17]	17.00	0.6040	0.10679	0.02425	0.56	256
DCPDN [18]	16.92	0.6098	0.10707	0.02511	0.78	256
MPRNet [19]	16.82	0.6252	0.10814	0.02515	15.74	128
SR3 [20]	16.54	0.6195	0.11237	0.02786	60.66	128
AECRNet [21]	16.53	0.5860	0.11107	0.02596	0.26	256
FFA-Net [22]	16.43	0.6003	0.11324	0.02639	1.97	256
GCANet [23]	15.91	0.5564	0.12777	0.03329	0.12	256
pix2pix [24]	15.58	0.6289	0.15443	0.03978	41.83	full
UNet++ [25]	15.05	0.6093	0.16398	0.04651	26.08	full
Ancuti fusion [26]	14.64	0.5025	0.14733	0.04322	0.07	256
DEA-Net [27]	14.22	0.5561	0.14530	0.04396	0.09	256
GridDehazeNet [28]	13.87	0.5400	0.17030	0.05093	0.96	256
DAT [29]	13.08	0.5157	0.17973	0.05701	0.43	full
DRCT [30]	12.70	0.5357	0.18905	0.06108	7.23	full

S4. FOG-CHAMBER CLASSIFICATION

The fog-chamber image pairs inherit labels from the archive categories used for display capture. We used these labels for a downstream semantic-preservation check, asking whether NAFNet restoration preserves category-recognition content for a classifier trained on clear images.

We excluded the illustration category and retained five categories: apparel, artwork, cars, dishes, and furniture. Illustrations overlapped visually with the artwork category, making the archive-label distinction poorly suited for this semantic-preservation check. The held-out test set used the same fog-chamber benchmark image IDs after this exclusion, giving 460 matched examples with 92 images per class. The remaining five-class paired image IDs were split into 3708 training and 412 validation examples with random seed 42.

We trained one torchvision ResNet-50 backbone [31] initialized from ImageNet weights on clear target images only and replaced the final layer with a five-way linear classifier. Training used 224×224 inputs, batch size 32, AdamW, and two stages: 5 frozen-backbone head epochs at learning rate 10^{-3} followed by up to 20 full-network fine-tuning epochs at learning rate 10^{-4} with cosine decay and early stopping by validation macro-F1. Mild augmentation used random resized crops, horizontal flips, and light color jitter.

The frozen clear-trained classifier was evaluated on three matched test inputs: clear targets, foggy inputs, and NAFNet-restored foggy inputs. The NAFNet-restored images were generated from the same fog-chamber benchmark checkpoint used for downstream fine-tuning. Table S3 reports the results.

Table S3. Five-class semantic-preservation check using one clear-trained ResNet-50 classifier. The classifier was trained on clear target images and evaluated without retraining on matched held-out clear targets, foggy inputs, and NAFNet-restored outputs.

Test input	n	Accuracy	Macro-F1	Top-2 accuracy
Clear targets	460	0.9891	0.9891	1.0000
Foggy inputs	460	0.8283	0.8346	0.9565
NAFNet-restored outputs	460	0.9413	0.9416	0.9783

NAFNet restoration improved recognition relative to foggy inputs for the frozen clear-trained classifier. On paired image IDs, restored outputs corrected 64 examples that were missed from foggy inputs and introduced 12 new errors. Against the paired clear targets, restored outputs remained below the clear-image ceiling.

S5. CLASSICAL DARK-CHANNEL-PRIOR BASELINE

To connect the controlled chamber images to the classical dehazing literature, we evaluated the dark channel prior of He, Sun, and Tang [32] on the same 552-image fog-chamber benchmark held-out split as a negative control. The implementation used the standard atmospheric scattering model with a 15-pixel dark-channel patch, atmospheric light estimated from the brightest pixel among the top 0.1% highest dark-channel responses, $\omega = 0.95$, guided-filter transmission refinement, and a minimum transmission floor of 0.10. Metrics were computed against the aligned archive targets at 512×512 resolution.

Table S4. Classical dark-channel-prior baseline on the 552-image benchmark held-out split. Input metrics compare the raw foggy image to the aligned clear target; DCP metrics compare the dark-channel-prior output to the same target.

Condition	n	Input PSNR	Input SSIM	DCP PSNR	DCP SSIM	DCP MAE	DCP MSE
Benchmark vertical-polarizer	552	11.80	0.5109	9.82	0.4364	0.2982	0.1386

The classical prior did not improve this laboratory display-imaging control. On the benchmark held-out split, DCP decreased mean PSNR from 11.80 dB for the raw foggy input to 9.82 dB and decreased SSIM from 0.5109 to 0.4364. This negative result is consistent with the method’s outdoor natural-image assumptions and supports using paired supervised restoration for the controlled fog-imager geometry.

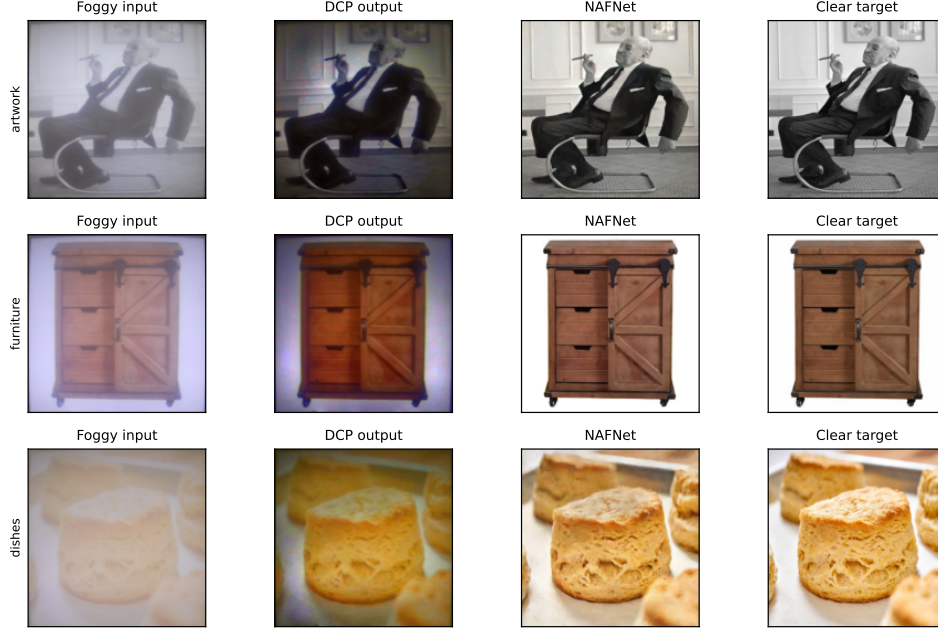


Fig. S1. Classical dark-channel-prior examples on the 552-image benchmark held-out split. Columns compare the foggy input, dark-channel-prior output, NAFNet output, and aligned clear target for representative archive categories.

S6. SYNTHETIC FINE-TUNING

The synthetic-fine-tuned model started from the fog-chamber NAFNet checkpoint and was fine-tuned on clear Mapillary Vistas images with synthetic fog generated on the fly. This fine-tuning branch used the local NAFNet wrapper in which the network output is added to the foggy input. Each sample used a spatially fogged crop as the input and the original clear crop as the target.

The synthetic image formation model used the atmospheric blend

$$\mathbf{I}(\mathbf{x}) = \mathbf{J}(\mathbf{x})T(\mathbf{x}) + \mathbf{A}(\mathbf{x})[1 - T(\mathbf{x})], \quad (\text{S5})$$

where $\mathbf{J}$ is the clear image, $\mathbf{A}$ is a spatially varying airlight field, and $T = \exp(-\beta d)$ is a synthetic transmission map. The depth-like map d mixed a geometric prior and smooth random depth map with weights 0.82 and 0.18. The attenuation β was modulated by a six-octave smooth random field with vertical and horizon-centered terms.

The evaluated depth-informed extension retained that normalized base map but used raw ZoeDepth output z from the Intel ZoeDepth NYU-KITTI model to modulate attenuation:

$$\beta(\mathbf{x}) = \text{clip}\{\beta_0[1 + 0.70z(\mathbf{x})], 0.03, 8\}.$$

The resulting transmission was $T(\mathbf{x}) = \exp[-\beta(\mathbf{x})d(\mathbf{x})]$. ZoeDepth maps were precomputed as float16 arrays and were not normalized per image. This branch is an evaluated generator extension rather than a new depth estimator or atmospheric model.

After the atmospheric blend, the generator added edge veil, airlight color variation, bloom, fog-coupled blur, partial desaturation, gamma adjustment, and weak additive noise. Airlight variation used independent smooth fields for the three color channels, and edge veil used a radial mask. During training, fog strength and spatial variation were sampled per crop. Light-fog crops further reduced the fog-strength and spatial-variation multipliers, and identity crops used the clear crop unchanged. Validation and test crops used deterministic center crops, fixed fog multipliers, and stable path-based seeds.

The generator parameter ranges were initialized from simple image statistics measured on free-space light-fog captures, including color means, luminance, contrast, and saturation. These captures were used only as a calibration reference for generator ranges; they were not used as synthetic-training, validation, or test images.

Figure S2 shows the training branches. The synthetic fine-tuning branch used fog-chamber pretraining followed by Mapillary synthetic-fog fine-tuning. The NH-HAZE, O-HAZE, and NTIRE branches were separate supervised task-specific fine-tuning checks.

Table S5. Selected synthetic-fog generator parameters. The scalar β is an internal synthetic-fog strength. Training sampled per-crop beta and spatial-variation multipliers around these base values.

Parameter	Value
base β mean	2.58
base β variation	0.94
fog-field scale	924 px
fog-field octaves	6
fog-field contrast	1.20
vertical gradient weight	0.25
horizon bias weight	0.20
airlight RGB	0.72, 0.715, 0.70
airlight spatial variation	0.045
bloom strength / radius	0.18 / 8 px
blur radius / fog coupling	0.9 px / 0.22
saturation mix	0.14
contrast gamma	1.00
edge-veil strength	0.10
noise strength	0.008

Table S6. Depth-informed generator settings. All training-budget and sampling settings matched the randomized branch.

Parameter	Value
depth source	Intel/zoedepth-nyu-kitti
base β mean / variation	0.59 / 0.55
fog-field scale / octaves	420 px / 3
airlight RGB / variation	0.72, 0.73, 0.76 / 0.05
bloom strength / radius	0.10 / 9 px
blur radius / fog coupling	0.9 px / 0.22
saturation mix / gamma	0.24 / 0.96
edge-veil / noise	0.245 / 0.008
raw-depth modulation weight	0.70

Matched evaluation used the same image order, crop policy, seed 734, effective batch size 2, and 2,600-update budget for both generators and all 30 architectures. Every randomized checkpoint improved on the fixed randomized-fog test (median +6.27 dB) and every depth-informed checkpoint improved on the fixed depth-fog test (median +5.13 dB). Depth-informed checkpoints were lower than randomized checkpoints on the randomized-fog test for all 30 models (median -2.08 dB), whereas they exceeded randomized checkpoints on the depth-fog test for all 30 models (median +3.41 dB). Both branches reduced chamber performance.

Five historical chamber checkpoints (DehazeFormer, DCPDN, GCANet, Ancuti fusion, and GridDehazeNet) were retrained because their files were no longer available. Table S2 remains the authoritative historical ranking; this new table supports matched within-architecture strategy comparisons.

Table S7. Selected synthetic fine-tuning settings. Validation and test PSNR are computed against synthetic Mapillary clear targets and are used only as synthetic-training diagnostics.

Setting	Value
clear-image source	Mapillary Vistas
model initialization	fog-chamber NAFNet checkpoint
split counts	18,000 train, 2,000 validation, 5,000 test
crop policy	random train crops; center validation/test crops
crop size	512×512
batch size	2
optimizer	AdamW
learning rate schedule	constant 10^{-4}
weight decay	10^{-4}
training budget	1 epoch capped at 2600 train batches
validation / test budget	300 / 500 batches
loss	L1 + residual TV
residual TV weight	0.008
mean-color loss weight	0.02
augmentation	horizontal flip 0.50, vertical flip 0.15
beta multiplier range	0.55–1.40
spatial-variation multiplier range	0.60–1.15
light-fog probability	0.30
identity probability	0.10
airlight jitter	± 0.03 per channel
seed policy	stable path hash; random train offset
randomized validation / test PSNR	21.56 / 21.61 dB
depth validation / test PSNR	19.65 / 19.86 dB

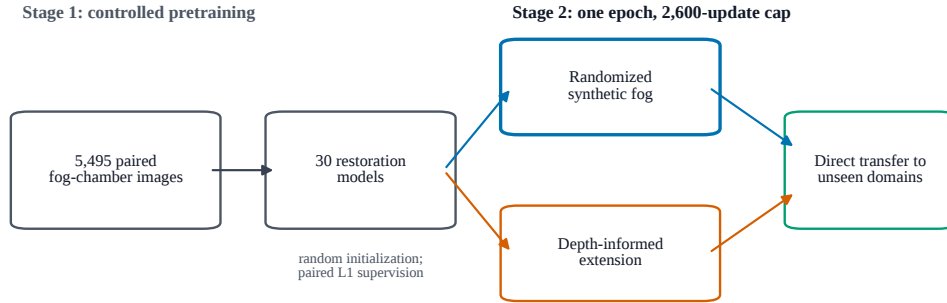


Fig. S2. Cross-domain transfer workflow. Paired fog-chamber data initialize the supervised NAFNet model. The synthetic fine-tuning branch renders Mapillary clear crops with spatial synthetic fog and fine-tunes against the original clear crops. The resulting model is applied to unpaired real-fog examples and paired public-haze/NTIRE examples without using those paired training splits. Separate NH-HAZE, O-HAZE, and NTIRE NAFNet models provide supervised task-specific checks.

Table S8. Matched PSNR comparison for all 30 restoration models. Each checkpoint is evaluated with the same architecture-specific wrapper on identical held-out inputs. Synthetic fine-tuning used one epoch capped at 2,600 optimizer updates. Values are mean PSNR in dB.

Model	Chamber test			Randomized-fog test			Depth-fog test		
	Chamber	Random.	Depth	Chamber	Random.	Depth	Chamber	Random.	Depth
NAFNet	24.26	14.74	15.80	12.36	21.74	16.79	12.57	16.55	19.87
SpecAT S2	22.99	13.56	13.42	12.56	19.76	15.59	12.50	13.40	18.47
RegGAN	22.50	14.01	13.77	10.96	17.79	16.60	11.05	16.58	16.81
ConvNeXt	22.39	13.09	13.22	11.83	21.12	16.70	12.32	14.03	19.58
RetinexFormer	21.76	14.12	14.33	13.24	20.42	16.99	13.09	13.96	18.63
HRNet	21.63	13.22	13.11	11.87	19.60	15.77	12.48	13.25	17.01
SpecAT S1	21.36	13.20	13.80	12.73	18.68	16.26	12.84	13.41	18.03
MST++	19.94	13.23	13.14	12.55	18.64	16.27	12.85	13.63	17.43
HDNet	19.74	12.79	12.00	12.99	18.14	16.58	12.91	13.16	15.39
PADUT	18.95	12.44	12.65	12.71	18.24	16.57	11.47	12.75	16.25
DehazeFormer	19.00	12.73	12.92	12.61	21.74	17.92	12.14	14.44	20.43
Swin2SR	18.35	12.09	12.29	12.29	21.03	17.41	11.69	14.41	18.04
RDN	18.17	11.89	11.40	12.48	17.84	16.23	11.72	12.68	16.60
Restormer	17.61	13.03	13.03	12.59	22.60	19.03	12.09	18.19	22.48
VmambaIR	17.56	11.67	11.96	11.96	17.18	16.15	10.43	12.64	14.07
MIRNet	17.11	12.15	12.00	12.42	19.61	17.77	12.06	14.09	19.72
MSBDN	16.99	11.89	11.94	12.60	18.88	16.34	11.42	13.36	16.65
DCPDN	16.63	12.60	12.60	12.97	20.12	16.96	11.50	14.09	18.26
MPRNet	16.80	11.42	9.08	12.28	19.83	18.12	11.56	15.13	18.39
SR3	16.52	12.05	12.31	12.22	21.91	18.29	11.81	15.89	20.83
AECRNet	16.52	12.06	12.31	12.69	20.03	17.61	11.87	14.68	18.89
FFA-Net	16.42	11.94	11.33	12.37	18.54	16.45	11.57	13.05	15.98
GCANet	15.74	11.49	11.64	12.21	17.61	15.69	11.57	13.16	15.81
pix2pix	15.57	12.61	12.45	12.14	15.58	15.17	11.17	12.87	14.00
UNet++	15.04	12.09	11.97	12.37	15.43	14.87	11.95	13.48	14.12
Ancuti fusion	14.57	11.23	10.89	12.62	16.84	15.90	11.27	12.97	14.95
DEA-Net	14.21	11.66	11.35	12.15	18.41	16.34	11.60	13.44	16.67
GridDehazeNet	14.02	11.21	11.19	11.81	15.70	14.51	11.16	13.38	14.82
DAT	13.07	12.17	11.32	11.23	14.43	14.32	9.29	12.81	13.51
DRCT	12.69	11.70	11.13	12.40	15.10	14.84	10.00	13.15	13.97

S7. PUBLIC PAIRED REAL-HAZE ADAPTATION

The main manuscript reports the mixed O-HAZE+NH-HAZE adaptation result. O-HAZE contains 45 paired outdoor real-haze images, and NH-HAZE contains 55 paired non-homogeneous real-haze images [33, 34]. The reported model used only the mixed training policy: O-HAZE images 1–35 and NH-HAZE images 1–45 for training, O-HAZE 36–40 and NH-HAZE 46–50 for validation, and O-HAZE 41–45 plus NH-HAZE 51–55 for testing.

The real-haze fine-tuning run started from the fog-chamber NAFNet model and used random 256×256 paired hazy/ground-truth crops, L1 reconstruction loss, AdamW, learning rate 10^{-4} , batch size 4, and early stopping by validation L1. The best validation epoch was 39, and training stopped after epoch 57.

We evaluated chamber, randomized-synthetic, and depth-informed NAFNet checkpoints on all available paired O-HAZE ($n = 45$), NH-HAZE ($n = 55$), and released NTIRE nighttime images ($n = 25$), without public-data training or model selection. Native-resolution inference used 1024-pixel tiles with 96-pixel overlap. The depth-informed checkpoint exceeded the randomized checkpoint by +1.49, +0.63, and +0.54 dB, respectively, with higher SSIM on all three datasets. Relative to the chamber checkpoint, depth-informed changes were +0.56, +0.67, and +1.78 dB.

Table S9. Supervised mixed real-haze fine-tuning protocol and split audit. Image identity was checked as dataset/image-id pairs across train, validation, and test records.

Model	Training images	Validation images	Test images	Best epoch	Overlap
Mixed real-haze fine-tuning	O-HAZE 1–35 + NH-HAZE 1–45 ($n=80$)	O-HAZE 36–40 + NH-HAZE 46–50 ($n=10$)	O-HAZE 41–45 + NH-HAZE 51–55 ($n=10$)	39	0 train/test; 0 validation/test

Table S10. Internal test-split metrics for the combined real-haze model. Subset rows are diagnostics of the same model.

Model	Held-out split	n	MAE	PSNR	SSIM
Mixed real haze	O-HAZE 41–45	5	0.0589	22.65	0.726
Mixed real haze	NH-HAZE 51–55	5	0.0894	18.77	0.640
Mixed real haze	O-HAZE 41–45 + NH-HAZE 51–55	10	0.0741	20.71	0.683

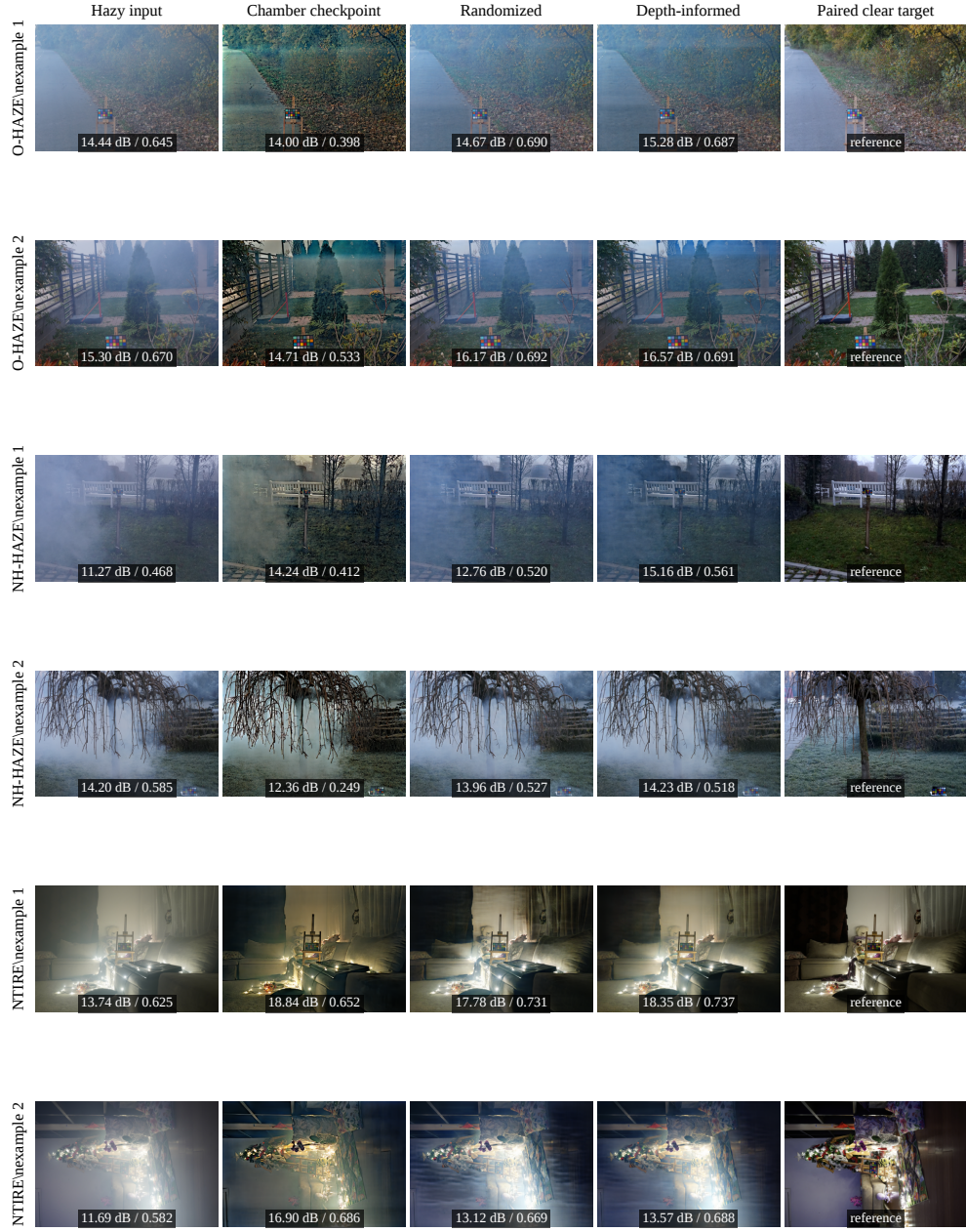


Fig. S3. Complete-dataset direct-transfer comparison. Columns show the hazy input, chamber checkpoint, randomized checkpoint, depth-informed checkpoint, and paired target. Summary metrics use all available paired images. The depth-informed extension exceeds the randomized branch on mean PSNR and SSIM for all three datasets.

S7.1. NTIRE 2026 Nighttime Haze Evaluation

The NTIRE 2026 evaluation used the released nighttime haze training pairs. The available training set contains 25 paired nighttime hazy/ground-truth images. We trained an NTIRE-specific NAFNet from the fog-chamber model using IDs 01–20 for training, IDs 21–23 for validation and model selection, and IDs 24–25 as an internal test split. The best validation epoch was 15, and training stopped after epoch 29. This internal split is useful for checking whether the same backbone adapts to nighttime paired haze, but it is not an official challenge validation/test result because those clear targets are unavailable locally. It therefore does not establish a competition rank.

Table S11. NTIRE 2026 supervised nighttime-haze result. The held-out IDs 24–25 were not used for training or model selection. Metrics are computed against the released training ground truth.

Split	Model	n	MAE	PSNR	SSIM
Held-out 24–25	NTIRE NAFNet	2	0.0435	24.77	0.770

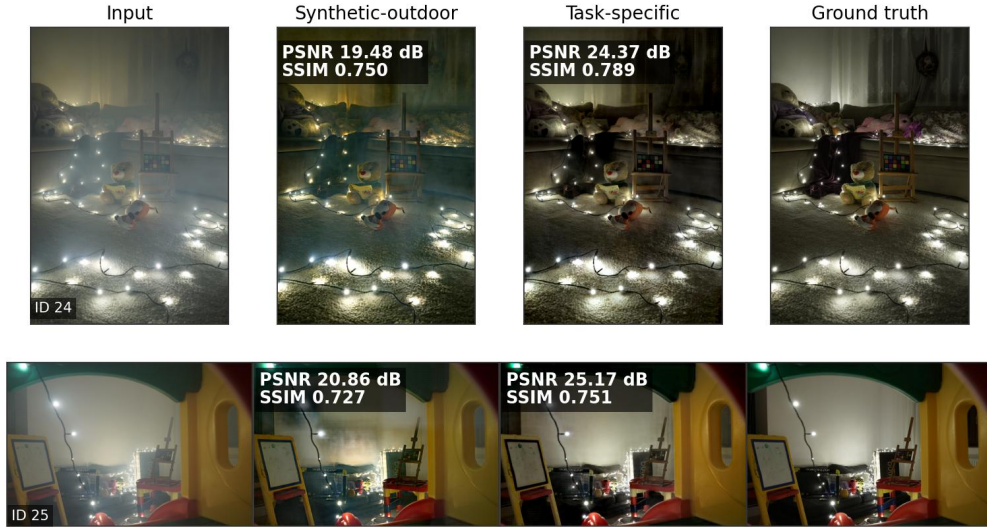


Fig. S4. NTIRE 2026 held-out examples 24 and 25 showing direct synthetic-fine-tuned transfer and task-specific supervised adaptation. Columns compare the hazy input, synthetic-fine-tuned NAFNet output with no NTIRE training, NTIRE-specific supervised NAFNet output, and paired ground truth. Output stamps on both model columns report PSNR/SSIM against the released ground truth.

The mixed real-haze result uses a 10-image internal test split, and the NTIRE 2026 check uses a two-image internal test split from the released training pairs. These checks show that the fog-chamber NAFNet model can be adapted by paired outdoor or nighttime haze supervision, but the training policy, split definition, image size, and reference appearance prevent leaderboard claims.

S8. FOG DENSITY STATISTICS

Fog statistics were computed from the 552-image fog-chamber benchmark held-out split and from the clear and light-fog outdoor image sets. Images were converted to numeric pixel values after resizing the longest side to at most 512 pixels for consistent throughput. The dark-channel haze index was the image mean of a 15-pixel minimum-filtered color-channel dark channel, RMS contrast was luminance standard deviation divided by mean luminance, sharpness was the mean finite-difference luminance gradient magnitude, and airlight shift was the change

in mean luminance relative to the available clear reference. Chamber statistics used paired fog/clear archive images, so contrast, luminance-standard-deviation, and gradient attenuation were reported as fog-over-clear ratios. The outdoor clear and light-fog captures are independent groups, so outdoor attenuation was reported only as a group-level light-fog/clear ratio.

Fog proxy statistics in chamber and outdoor data

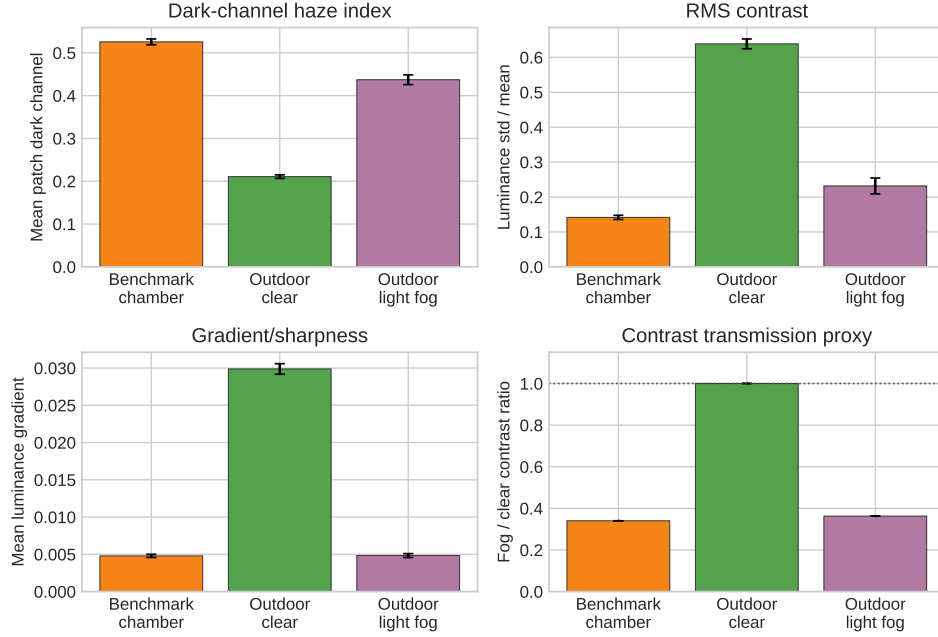


Fig. S5. Simple fog statistics for the chamber and outdoor data. The chamber condition uses paired fog/clear archive references; the outdoor light-fog condition uses the outdoor clear group as an independent reference. Error bars show 95% confidence intervals for absolute per-image metrics.

Table S12. Fog statistics comparing chamber and outdoor data. Ratios are fog/reference values, where the reference is the paired clear archive image for chamber data and the outdoor clear group mean for outdoor light fog.

Condition	n	Dark channel	RMS contrast	Grad. mag.	Contrast ratio
Benchmark chamber	552	0.526	0.142	0.00478	0.340
Outdoor clear	430	0.211	0.639	0.02988	1.000
Outdoor light fog	84	0.437	0.232	0.00483	0.363

The statistics separated clear and fogged imagery in both acquisition regimes. For the benchmark chamber data, RMS contrast was 0.340 of the paired clear reference and mean gradient was 0.181 of clear. Outdoor light fog also moved in the expected direction relative to outdoor clear: dark-channel haze index increased from 0.211 to 0.437, RMS contrast fell to 0.363 of the clear outdoor group, and mean gradient fell to 0.162 of clear. Thus the statistics support a defensible relative fog-index comparison across chamber and free-space data, but they do not establish physical fog density without distance and calibration.

S9. IMAGE-DOMAIN POWER-SPECTRAL ANALYSIS

We used the paired chamber data to quantify how fog changes spatial-frequency content in the recorded images. For each held-out pair, the foggy chamber capture and the correspond-

ing archive image displayed on the screen were converted to luminance, resized to 512×512 , mean-subtracted, Hann-windowed, Fourier transformed, and radially averaged into 96 spatial-frequency bins. We then computed the per-pair radial power-spectral-density (PSD) ratio,

$$R_{\text{PSD}}(f) = \frac{P_{\text{fog}}(f)}{P_{\text{archive}}(f)}, \quad (\text{S6})$$

where $P_{\text{fog}}(f)$ is the foggy camera-capture PSD and $P_{\text{archive}}(f)$ is the PSD of the displayed archive image. The deterministic every-tenth held-out split gave $n = 552$ pairs.

This is an image-domain degradation measurement, not a physical transmission spectrum of the fog. The ratio includes the display, chamber, fog, lens, sensor, exposure, camera processing, and resizing pipeline. It is therefore used only to summarize loss of recorded spatial detail.

The median paired PSD ratio was 0.0322 (95% bootstrap CI 0.0294–0.0372) over low spatial frequencies, 0.01–0.05 cycles/pixel, and 0.0036 (95% CI 0.0033–0.0041) over higher spatial frequencies, 0.15–0.24 cycles/pixel. Thus high-frequency power was retained $8.9\times$ less than low-frequency power in the paired image-domain measurement. This frequency-domain result supports the Laplacian-retention analysis in the main text: foggy captures lose fine spatial structure disproportionately, making high-texture scenes harder to restore.

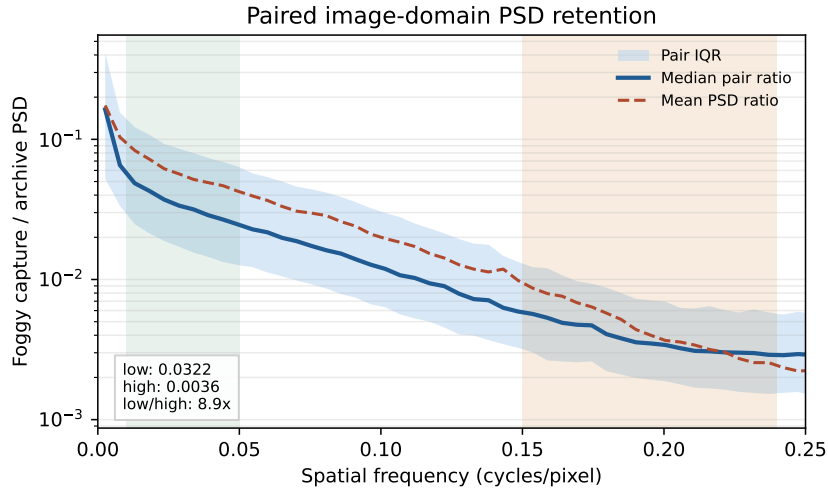


Fig. S6. Paired image-domain power-spectral-density retention for the held-out chamber split. The solid line shows the median foggy-capture/archive PSD ratio across image pairs; the shaded band shows the interquartile range. The dashed line shows the ratio of mean foggy and archive PSDs. Green and orange bands mark the low- and high-frequency ranges summarized in the text.

S10. PAIRED STRUCTURE-LOSS METRICS AND RESTORATION DIFFICULTY

We used the 552-image fog-chamber benchmark held-out split to ask whether simple image statistics tracked image-level NAFNet restoration quality. Each foggy image was paired with its aligned clear archive target. We compared foggy-image-only haze proxies with paired structure-loss metrics computed from the fog/clear pair.

The strongest single paired statistic was the mean absolute Laplacian-response ratio,

$$r_{\nabla^2} = \frac{\text{mean}(|\nabla^2 I_{\text{fog}}|)}{\text{mean}(|\nabla^2 I_{\text{clear}}|)}, \quad (\text{S7})$$

where I_{fog} and I_{clear} are luminance images. This ratio measures how much fine spatial structure remains in the foggy image relative to the aligned clear target. It had Spearman correlation $\rho = 0.632$ with per-image NAFNet PSNR. By comparison, the strongest foggy-image-only statistic in this subset, the 90th percentile dark-channel response, reached $\rho = 0.399$.

We also evaluated the same mean-absolute Laplacian ratio using wider finite-difference steps, where step 1 is the nearest-neighbor five-point Laplacian and larger steps compare pixels separated by 3, 8, or 12 pixels. Correlation with NAFNet PSNR decreased as the step widened ($\rho = 0.542$ at step 3 and 0.417 at step 8), indicating that the most predictive signal was fine-scale structure retention rather than broader luminance variation. A separate PSD fit summarized each fog/clear pair by a single equivalent Gaussian-PSF width, but this scalar blur width was only a weak predictor of restoration quality ($\rho = -0.206$), so we treat it as a diagnostic rather than the primary difficulty metric.

Paired structure loss tracks NAFNet restoration difficulty

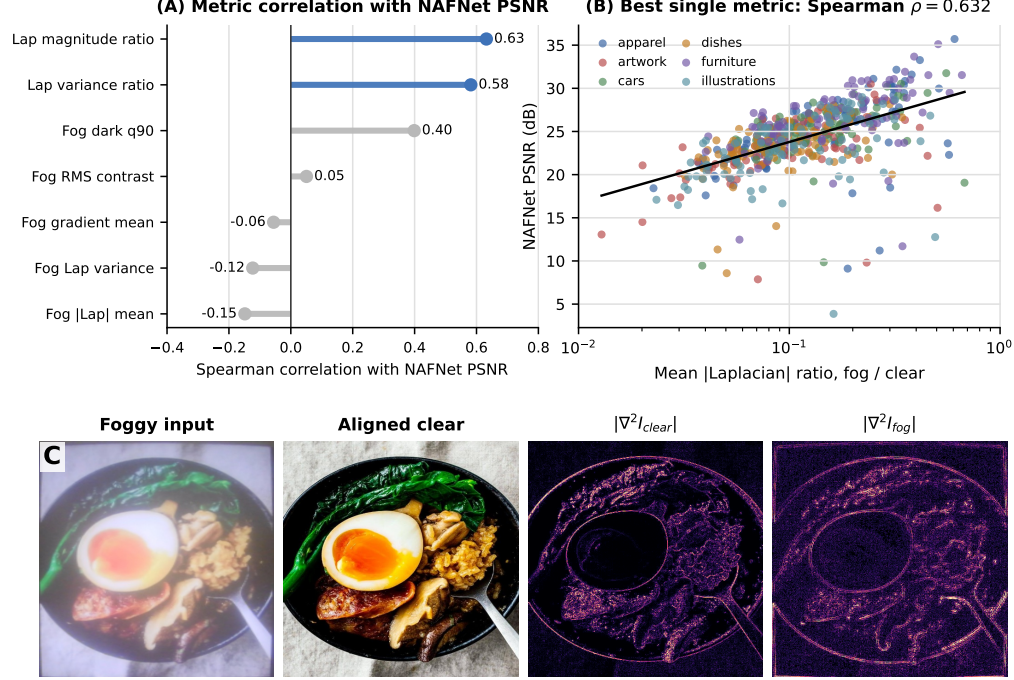


Fig. S7. Paired structure-loss metrics track NAFNet restoration difficulty on the fog-chamber held-out split. (A) Signed Spearman correlations between selected paired and foggy-only image statistics and per-image NAFNet PSNR, with ρ values labeled next to each point. (B) Per-image NAFNet PSNR plotted against the strongest single metric, the scalar fog/clear mean absolute Laplacian-response ratio. Points are colored by archive category, and the black line shows a fit versus log₁₀ of the metric. (C) Example showing the foggy input, aligned clear target, and independently contrast-scaled absolute Laplacian-response maps for clear and foggy images.

The statistic is a scalar ratio of two full-image averages, not the average of a pixelwise ratio map. Thus, locations with larger absolute Laplacian response contribute more to the numerator and denominator, whereas flat regions contribute little. These results suggest that image-level restoration difficulty is not captured by foggy-image haze proxies alone. Paired structure loss, especially loss of Laplacian response, better tracks NAFNet PSNR because it measures how much high-frequency scene detail survives the fog-imaging process. The statistic is a relative restoration-difficulty index, not a calibrated physical fog-density measurement.

Table S13. No-reference NIQE scores for representative out-of-domain examples. Lower NIQE is better. The fog-machine and aircraft-window rows use the synthetic-fine-tuned output images. The O-HAZE and NH-HAZE rows use the mixed real-haze supervised output images, and the NTIRE row uses the NTIRE-specific supervised output images.

Dataset	n	Input NIQE	Output NIQE	NIQE decrease
Aircraft-window	43	6.22	4.97	1.25
Fog-machine free fog	99	5.05	3.61	1.44
O-HAZE	5	4.15	4.03	0.12
NH-HAZE	5	2.83	3.55	-0.71
NTIRE 2026 nighttime haze	2	4.94	4.84	0.10

S11. SYNTHETIC-CAMERA AND IMAGE-SIGNAL-PROCESSING SENSITIVITY

We performed a controlled sensitivity study to separate camera and image-signal-processing effects from changes in scene content. The study used 96 images from the held-out fog-chamber split, balanced as 16 images from each of six archive categories. No model parameters were trained or selected on these images. Each foggy input x was evaluated under 43 conditions: an unchanged identity condition and 42 nonidentity settings spanning exposure, white balance, tone response, Gaussian blur, JPEG compression, sensor shot/read noise, resampling, and combined profiles. The same deterministic setting and stochastic noise basis were applied to the foggy input, its aligned clear target, and the baseline model output.

For a fixed model f and camera operator C , input-output tracking was measured from

$$d_{\text{in}}(C) = \text{PSNR}(x, C(x)), \quad d_{\text{out}}(C) = \text{PSNR}(f(x), f(C(x))). \quad (\text{S8})$$

Because larger PSNR indicates a smaller perturbation, a positive correlation between d_{in} and d_{out} indicates that conditions producing larger input changes tend to produce larger output changes. Tracking alone does not establish restoration accuracy. We therefore used unchanged-target accuracy,

$$A_{\text{canonical}}(C) = \text{PSNR}(f(C(x)), y), \quad (\text{S9})$$

where y is the aligned clear target, as the primary robustness measure. Confidence intervals for condition means used image bootstrap resampling; tracking intervals used 1,000 clustered bootstrap resamples that retained each selected image’s complete condition set.

The correctly loaded chamber-trained NAFNet reached 24.000 dB PSNR and 0.7791 SSIM in the identity condition on this 96-image subset, consistent with its 24.33 dB mean on the complete 552-image benchmark split. Across the 42 nonidentity condition means, input and output changes were positively associated (Pearson $r = 0.706$, 95% clustered-bootstrap CI 0.693–0.719; Spearman $\rho = 0.665$). The fitted slope was 0.451 (95% CI 0.439–0.463), so the curves tracked in ordering but not one-to-one in decibels. Ordering within individual perturbation families was stronger ($r = 0.973$ –0.999), showing that severity produced coherent curves even though response magnitude differed among camera operations.

Camera and processing changes also reduced reconstruction accuracy. The largest tested failure was the strongest sensor-noise condition, which reduced canonical-target PSNR by 12.70 dB on average (95% CI 12.04–13.36 dB loss). Large exposure and white-balance shifts and the combined profiles also caused substantial losses, whereas the tested resampling range had little effect. Figure S8 combines the requested tracking analysis with the corresponding unchanged-target accuracy curves.

These perturbations are controlled abstractions rather than a physical camera model. They do not reproduce demosaicing, rolling shutter, autofocus, flare, lens-specific blur, proprietary denoising, high-dynamic-range fusion, or all interactions among sensor and processing stages. Consequently, this experiment identifies model sensitivities and supports the bounded claim that input and output changes track, but it is not a substitute for paired clear/fog capture using a physically different camera.

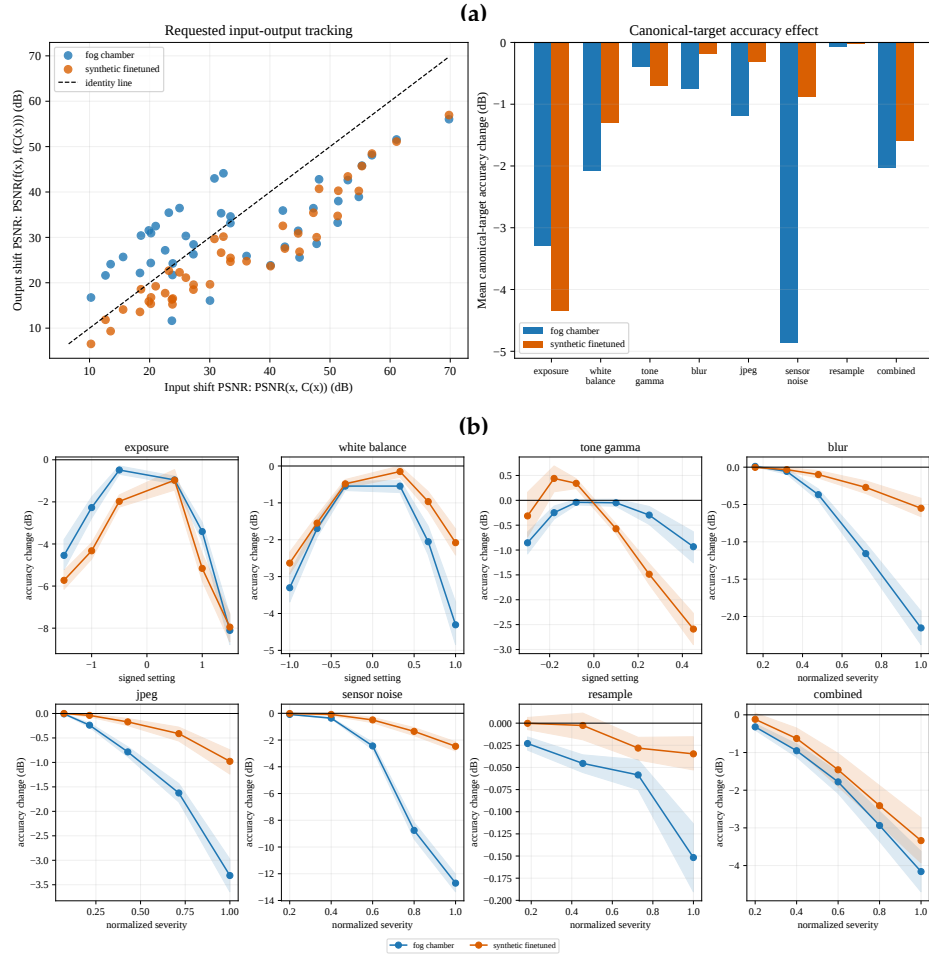


Fig. S8. Controlled synthetic-camera sensitivity of fixed NAFNet checkpoints. (a) Left: condition-mean input-shift versus output-shift PSNR for the 42 nonidentity settings; the dashed line denotes equal input and output change, not a regression fit. Right: mean canonical-target accuracy change grouped by perturbation family. (b) Canonical-target accuracy change as a function of signed setting or normalized severity; shaded bands are 95% image-bootstrap confidence intervals. Blue denotes the correctly loaded fog-chamber checkpoint. Orange denotes the archived randomized-synthetic checkpoint as a descriptive sensitivity probe only: its historical training path changed the output wrapper, so the two curves do not constitute a controlled before/after fine-tuning comparison. Both panels are vector graphics with embedded fonts.

REFERENCES

1. Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Transactions on Image Process.* **13**, 600–612 (2004).
2. L. Chen, X. Chu, X. Zhang, and J. Sun, "Simple baselines for image restoration," in *European Conference on Computer Vision*, (2022), pp. 17–33.
3. Z. Yao, S. Liu, X. Yuan, and L. Fang, "Specat: Spatial-spectral cumulative-attention transformer for high-resolution hyperspectral image reconstruction," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, (2024), pp. 25368–25377.
4. L. Kong, C. Lian, Z. Li, *et al.*, "Breaking the dilemma of medical image-to-image translation," in *Advances in Neural Information Processing Systems*, vol. 34 (2021).
5. Z. Liu, H. Mao, C.-Y. Wu, *et al.*, "A convnet for the 2020s," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, (2022), pp. 11976–11986.
6. Y. Cai, H. Bian, J. Lin, *et al.*, "Retinexformer: One-stage retinex-based transformer for low-light image enhancement," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, (2023), pp. 12504–12513.
7. J. Wang, K. Sun, T. Cheng, *et al.*, "Deep high-resolution representation learning for visual recognition," *IEEE Transactions on Pattern Analysis Mach. Intell.* **43**, 3349–3364 (2020).
8. Y. Cai, J. Lin, X. Hu, *et al.*, "Mask-guided spectral-wise transformer for efficient hyperspectral image reconstruction," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, (2022), pp. 360–369.
9. X. Hu, Y. Cai, J. Lin, *et al.*, "Hdnet: High-resolution dual-domain learning for spectral compressive imaging," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, (2022), pp. 17542–17551.
10. M. Li, Y. Fu, J. Liu, and Y. Zhang, "Pixel adaptive deep unfolding transformer for hyperspectral image reconstruction," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, (2023), pp. 12902–12911.
11. Y. Song, Z. He, H. Qian, and X. Du, "Vision transformers for single image dehazing," *IEEE Transactions on Image Process.* **32**, 1927–1941 (2023).
12. M. V. Conde, U.-J. Choi, M. Burchi, and R. Timofte, "Swin2sr: Swinv2 transformer for compressed image super-resolution and restoration," in *European Conference on Computer Vision Workshops*, (2022), pp. 669–687.
13. Y. Zhang, Y. Tian, Y. Kong, *et al.*, "Residual dense network for image super-resolution," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, (2018), pp. 2472–2481.
14. S. W. Zamir, A. Arora, S. Khan, *et al.*, "Restormer: Efficient transformer for high-resolution image restoration," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, (2022), pp. 5728–5739.
15. Y. Shi, B. Xia, X. Jin, *et al.*, "Vmambair: Visual state space model for image restoration," *arXiv preprint arXiv:2403.11423* (2024).
16. S. W. Zamir, A. Arora, S. Khan, *et al.*, "Learning enriched features for real image restoration and enhancement," in *European Conference on Computer Vision*, (2020), pp. 492–511.
17. H. Dong, J. Pan, L. Xiang, *et al.*, "Multi-scale boosted dehazing network with dense feature fusion," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, (2020), pp. 2157–2167.
18. H. Zhang and V. M. Patel, "Densely connected pyramid dehazing network," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, (2018), pp. 3194–3203.
19. S. W. Zamir, A. Arora, S. Khan, *et al.*, "Multi-stage progressive image restoration," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, (2021), pp. 14821–14831.
20. C. Saharia, J. Ho, W. Chan, *et al.*, "Image super-resolution via iterative refinement," *IEEE Transactions on Pattern Analysis Mach. Intell.* **45**, 4713–4726 (2023).
21. H. Wu, Y. Qu, S. Lin, *et al.*, "Contrastive learning for compact single image dehazing," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, (2021), pp. 10551–10560.
22. X. Qin, Z. Wang, Y. Bai, *et al.*, "Ffa-net: Feature fusion attention network for single image dehazing," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34 (2020), pp. 11908–11915.
23. D. Chen, M. He, Q. Fan, *et al.*, "Gated context aggregation network for image dehazing

- and deraining,” in *IEEE/CVF Winter Conference on Applications of Computer Vision*, (2019), pp. 1375–1383.
24. P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, “Image-to-image translation with conditional adversarial networks,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, (2017), pp. 1125–1134.
 25. Z. Zhou, M. M. Rahman Siddiquee, N. Tajbakhsh, and J. Liang, “Unet++: A nested u-net architecture for medical image segmentation,” in *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*, (2018), pp. 3–11.
 26. C. O. Ancuti and C. Ancuti, “Single image dehazing by multi-scale fusion,” *IEEE Transactions on Image Process.* **22**, 3271–3282 (2013).
 27. Z. Chen, Y. Wang, Y. Yang, and D. Liu, “Dea-net: Single image dehazing based on detail-enhanced convolution and content-guided attention,” *IEEE Transactions on Image Process.* **33**, 1002–1015 (2024).
 28. X. Liu, Y. Ma, Z. Shi, and J. Chen, “Griddehazenet: Attention-based multi-scale network for image dehazing,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, (2019), pp. 7314–7323.
 29. Z. Chen, Y. Zhang, J. Gu, *et al.*, “Dual aggregation transformer for image super-resolution,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, (2023), pp. 12312–12321.
 30. C.-C. Hsu, C.-M. Lee, and Y.-S. Chou, “Drct: Saving image super-resolution away from information bottleneck,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, (2024).
 31. K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, (2016), pp. 770–778.
 32. K. He, J. Sun, and X. Tang, “Single image haze removal using dark channel prior,” *IEEE Transactions on Pattern Analysis Mach. Intell.* **33**, 2341–2353 (2011).
 33. C. O. Ancuti, C. Ancuti, C. De Vleeschouwer, and P. Bekaert, “O-haze: A dehazing benchmark with real hazy and haze-free outdoor images,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, (2018).
 34. C. O. Ancuti, C. Ancuti, and R. Timofte, “Nh-haze: An image dehazing benchmark with non-homogeneous hazy and haze-free images,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, (2020).